

Drowning-Detecting Wristband Project Implementation Plan

Drowning remains a significant public health concern globally, claiming the lives of 236,000 people per year (WHO, 2023). Drowning also ranks as the third leading cause of unintentional injury death, accounting for 7% of all injury-related casualties, and impacts young children more than any other demographic, with children between the ages of 1 - 9 having the highest drowning rates (Stop Drowning Now, 2020). The current approach for ensuring the safety of swimmers is to rely on life guards. Lifeguards are trained to respond to drowning situations as quickly as possible, but being human, they may not recognize dangers quickly enough. A study from the Singapore Medical Journal found that 867% of drownings in public pools happened in the presence of a lifeguard (Chan, 2018). The truth about drowning, however, is that it is largely preventable, and drowning incidents can be further mitigated through the incorporation of technology in the field of water safety.

The objective of our project is to build a wristband that is capable of detecting drowning, especially at the early stages of the drowning process. The wristband contains several sensors, including a gyroscope, accelerometer, and pulse sensor. Data will be gathered from these sensors, and will then be processed to determine if the wearer is drowning. Once drowning is detected, a signal is transmitted from the wristband to a computer at the lifeguard's station, enabling the lifeguard to take immediate action. The conventional approach for drowning-detecting technology relies on the amount of time the wearer spends under a certain depth threshold. The issue with this approach, however, is that drowning is only detected once the wearer has taken in a dangerous amount of water and has reached the point of unconsciousness. This can be extremely problematic as the difference between survival and fatality can be a matter of seconds, making early detection vital. It is important to note that the intention of our system is not to replace conventional lifeguards, but rather to serve as an added layer of protection and security.

Our approach for drowning detection relies on the implementation of an artificial intelligence model that is trained on data of safe swimming activity, which will be collected through use of the wristband by a multitude of people, providing a diverse data pool. This will allow the wristband to build a profile of what safe swimming activity is like, and enables the system to detect deviations that would indicate distress. The model will also be built to recognize early and direct signs of drowning, such as a significant increase in heart rate from baseline level, as well as erratic movements detected by the gyroscope and accelerometer.

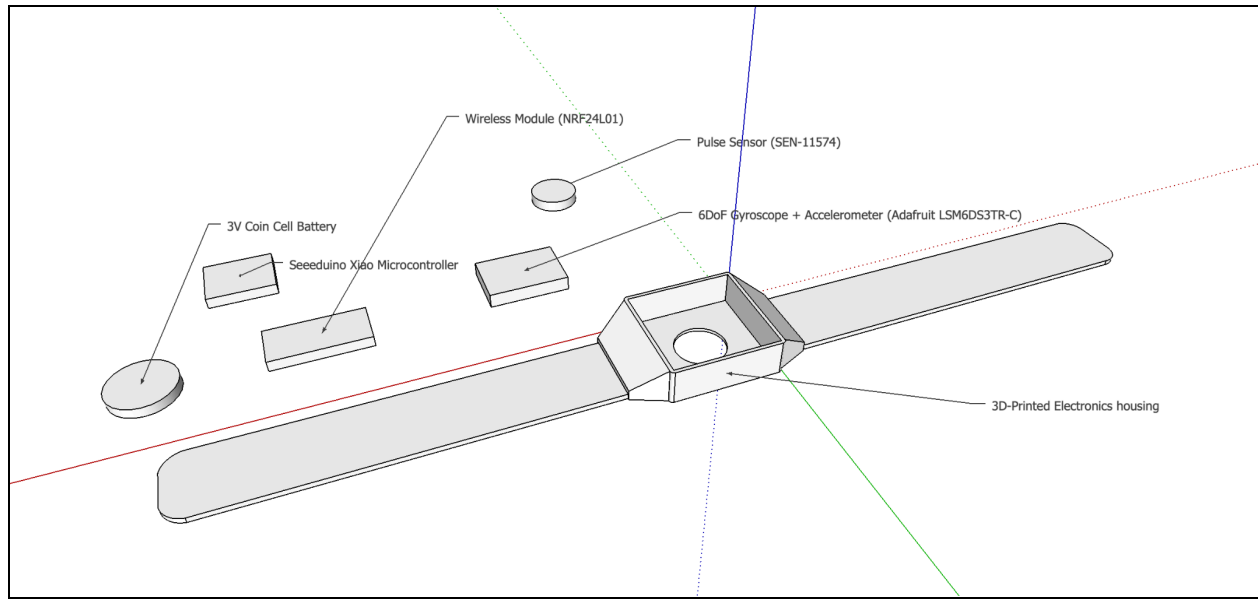


Figure 1: 3D model of the physical components of the drowning detection wristband

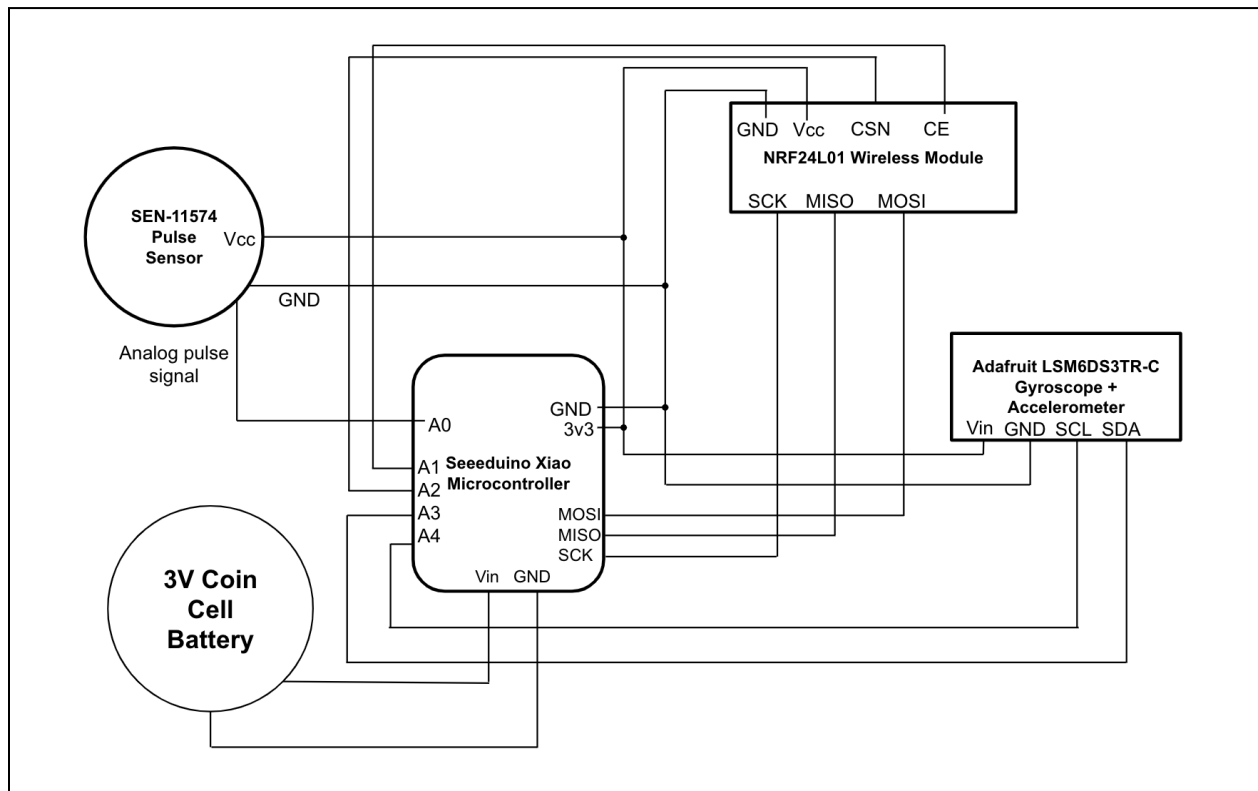


Figure 2: Circuit diagram of the drowning detection system

Materials and Equipment

Electronic Components:

1. Seeeduino Xiao microcontroller
2. NRF24L01 transceiver module
3. Adafruit LSM6DS3TR-C 6DoF Gyroscope + accelerometer
4. 3V coin cell battery
5. SEN-11574 pulse sensor

Additional Materials:

1. PETG 3D printing filament
2. TPU 3D printing filament
3. Epoxy resin
4. Soldering wire

Equipment:

1. Soldering iron
2. FDM 3D printer
3. Multimeter

Project Implementation Schedule

[illegible]

Team Member Roles

Afif Zughayer: In charge of hardware.

Kareem Mahmoud: in charge of software and coding.

Omar Aloruk: Incharge of 3D printing and fabrication.

Rayyan Agha Alnimer: In charge of data processing.

Sponsors and Marketing Plan

Potential sponsorship partners could be the police. Since drowning is a serious public safety risk, getting entities such as Dubai or Abu Dhabi Police involved could benefit in ensuring the effectiveness and reliability of our technology. We would market our product to institutions in charge of swimming areas, both public and private. What comes to mind are UAE Municipalities, private swimming pools/fitness institutions, and water parks. Since many families and children visit these places, these institutions would be concerned with the safety of their consumers given that young children are the demographic at greatest risk of drowning.

References

Chan, J. (2018). Drowning in swimming pools: Clinical features and safety recommendations based on a study of descriptive records by emergency medical services attending to 995 calls. *Singapore Medical Journal*, 59(1), 44-49. <https://doi.org/10.11622/smedj.2017021>

Stop Drowning Now. (2020, August 27). *Facts & stats about drowning*.
<https://stopdrowningnow.org/drowning-statistics/>

WHO. (2023, July 25). *World Drowning Prevention Day 2023*. World Health Organization.
<https://www.who.int/campaigns/world-drowning-prevention-day/2023>